POSTGRESQL · GRADE 11 · HOMEWORK

Your assignment: what, where and how

One assignment per student, drawn at random · **due in one week, before the next lesson** · handed in on GitHub

This sheet online: tinyurl.com/tues-hw-guide repo: tues-postgres

1. Find your assignment

You were given a number from 1 to 8. Open the link for your number (type it or scan it from the sheet). The page has everything: the story, the questions your database must answer, the minimum data, what to hand in and how.

#	ASSIGNMENT	DATABASE	LINK
1	The Class Bookshelf	`books`	tinyurl.com/tues-hw-1
2	Group Food Order	`food`	tinyurl.com/tues-hw-2
3	Playlists	`playlists`	tinyurl.com/tues-hw-3
4	Inter-class Tournament	`tournament`	tinyurl.com/tues-hw-4
5	Electronic Gradebook	`gradebook`	tinyurl.com/tues-hw-5
6	Weekly Timetable	`timetable`	tinyurl.com/tues-hw-6
7	Discord-style Chat	`chat`	tinyurl.com/tues-hw-7
8	Classroom Inventory	`inventory`	tinyurl.com/tues-hw-08

AI is allowed. Use whatever helps. The one rule: next lesson you show your schema and explain every table and column yourself. Then I change the requirements a little and you adapt the schema on the spot. If you cannot say why something is there, the work is not yours.

What I check: the dump restores without errors · every table has a PRIMARY KEY · relationships use FOREIGN KEY · required columns are NOT NULL · sensible types (DATE for dates, NUMERIC for money) · both queries return the right answer · the README explains why.

2. Get GitHub ready (once, for the whole year)

1. Go to `github.com` → **Sign up**. Any e-mail works. Pick a username you will not be embarrassed by – it stays on your work.
2. Top right **+** → **New repository**. Name: `tues-postgres`. Visibility: **Public**. Tick **Add a README file**. Click **Create repository**.
3. Give me your GitHub username in class (I keep a list). Your repository address is `github.com/USERNAME/tues-postgres`.

Do the work in pgAdmin

1. Start Docker (Docker Desktop, or `docker compose up -d` in your lesson-1 folder). pgAdmin: `localhost:5050`, password `postgres`.
2. Create a database with the name from your sheet. In it: **Query Tool** → `CREATE TABLE`, `INSERT`, then the two `SELECT`s.
3. Write `README.md` (3–5 sentences: the decisions you made and why) and `queries.sql` (the two queries with a comment above each).

3. Export the database and upload the three files

In a terminal, with Docker running (replace `DBNAME` with the name from your sheet):

```
docker exec local-postgres pg_dump -U postgres -d
DBNAME -f /tmp/dump.sql
docker cp local-postgres:/tmp/dump.sql dump.sql
```

`dump.sql` appears in the folder where you ran the commands.
Open it in a text editor: you should see your `CREATE TABLE`s and `INSERT` / `COPY` lines with your data.

1. In your repository on GitHub: **Add file** → **Create new file**. In the name box type `assignment-N/README.md` (your number instead of N – the slash creates the folder). Paste your text → **Commit changes**.
2. Open the folder `assignment-N` → **Add file** → **Upload files** → drag in `dump.sql` and `queries.sql` → **Commit changes**.
3. Check in the browser: `github.com/USERNAME/tues-postgres/tree/main/assignment-N` shows all three files. If you know `git`, clone/commit/push works just as well.